Does a personal closet make a better stylist?

Research pilot · October 19, 2026

Closet Quest helps people discover wearable combinations from clothes they already own. This evaluation tests two narrower claims: whether one photo can yield several useful wardrobe entries, and whether preferences, memory and retrieved notes improve recommendation relevance.

Public photos	Initial styling study	Human participants
18 photos / 36 initial calls	6 briefs / 18 generation runs	0 — human outcomes unmeasured

Evaluation design

Recognition used Fashionpedia validation annotations [1,2], mapped in advance to the app's six garment categories. From 127 eligible images with CC BY 2.0 or public-domain metadata and at least two categories, a fixed SHA-256 ranking selected 12 general images and six disjoint hat examples. The manifest was written before predictions. This is a small, enriched sample, not a population estimate or a model-training holdout.

Styling used 11 frozen wardrobe records and six fictional briefs covering campus, coffee and presentations. Every condition received the same explicit brief and owned pieces. Generation order rotated across briefs. The full condition additionally received fictional saved preferences, one feedback record and a personal note indexed in SQLite with OpenAI embeddings.

Condition	Information and execution
Quick baseline	Deterministic planner; lexical wardrobe retrieval; personal preferences, feedback and document retrieval disabled.
OpenAI, no memory	Same OpenAI generator and dense wardrobe retrieval as full; personal preferences, feedback and documents disabled.
Full personal RAG	OpenAI generator plus explicit preferences, feedback scoring and hybrid document retrieval.

Independent model feedback, not independent human evidence

GPT-6 Astra [3] rated anonymized X/Y/Z sets twice per brief, reversing presentation order. It scored brief fit, preference fit, coherence and evidence usefulness from 1 to 5, plus whether at least one look was acceptable under its rubric. There were 12 judge calls and 36 set ratings. The judge saw garment metadata and retrieved passages, not rendered outfits or actual wearability.

The no-memory comparison tests the combined personalization layer. It does not isolate the causal effect of document retrieval from explicit color filters or feedback. Scenarios were authored for this feature; they are not an independent user sample.

A second, controlled 2x2 experiment (page 4) separates documents from feedback memory across six new authored briefs, with 24 initial calls and six immediate repeats. Its predefined target-combination score uses garment IDs, not an AI judge.

More pieces found.
More work still needed.

RECOGNITION / FROZEN INITIAL BENCHMARK

Metric	Single-piece baseline	Multi-piece pipeline
Completed attempts	17/18	16/18
Category precision	17/17 (100.0%)	41/45 (91.1%)
Category recall	17/49 (34.7%)	41/49 (83.7%)
Exact category set	0/18 (0.0%)	12/18 (66.7%)
Hat detection, enriched subset	0/6 (0.0%)	6/6 (100.0%)
Median attempt latency	3.81 s	9.44 s

What improved

The multi-piece flow recovered 24 additional correct category occurrences: recall rose from 34.7% to 83.7% on the combined sample. All six annotated hats were detected, compared with none in the dominant-item baseline. On the 12-image general stratum alone, recall was 22/30 (73.3%), versus 11/30 (36.7%). The stronger hat result should not obscure the weaker general result.

What the numbers mean

A correct category contributes one true positive per image, even if several instances share that category. Precision = correct predicted categories / all predicted categories; recall = correct predicted categories / all annotated categories. Failed requests predict an empty set and remain in the recall and attempt denominators. Hats are scored separately by the returned item type. This evaluates inventory coverage, not bounding-box IoU, fabric, color, or fine-grained garment accuracy.

Both flows used GPT-6 Luna. Multi-piece mode also uses higher image detail and a larger output allowance, so this is a workflow comparison, not an isolated prompt experiment. Two workers ran concurrently. Timings cover direct local pipeline execution and provider calls, including failed attempts; they are neither phone-to-screen times nor hosted cold-start measurements.

Failure analysis and a targeted correction

The original multi-piece flow rejected images 15707 and 9711 with AMBIGUOUS_ITEM, losing clear detections alongside an uncertain piece. Version 1.1 now omits uncertain pieces individually and shows an explicit review warning; malformed geometry still fails validation. A separate, targeted two-image recheck recovered image 15707 with 4/4 categories. Image 9711 still failed with INVALID_SCENE_RESPONSE. This 1/2 recheck is not a replacement accuracy score: it was selected after observing errors and model outputs are stochastic.

Four extra predicted categories disagree with the dataset labels. They require visual adjudication before calling them hallucinations; coarse mapping and annotation omissions can also create disagreement. No phone-photo accuracy or human correction-time claim is supported yet.

Personal relevance improved in a synthetic test.

STYLING / SAME BRIEFS, THREE CONDITIONS

Metric	Quick	No memory	Full RAG
Completed attempts	6/6	6/6	6/6
Valid owned outfits	18/18 (100.0%)	14/14 (100.0%)	9/9 (100.0%)
Median pipeline latency	4 ms	4.49 s	3.91 s
Model fallback / attempts	0/6 (0.0%)	0/6 (0.0%)	0/6 (0.0%)
Acceptable in BOTH judge orders	1/6 (16.7%)	2/6 (33.3%)	6/6 (100.0%)
Preference fit / 5	2.58	2.67	4.67
Brief fit / 5	3.67	3.75	4.92
Coherence / 5	4.00	3.83	4.67

The measured signal

Full RAG received an acceptable rating in both orderings for 6/6 briefs; the quick baseline did so for 1/6 and OpenAI without memory for 2/6. Across the two ratings per brief, preference fit averaged 4.67/5 for full RAG, 2.58 for quick and 2.67 for no memory. There was no binary order disagreement in this small sample. Repeated ratings of the same six cases are not 12 independent users.

All 41 returned outfits passed owned-item and completeness checks. Full RAG returned only nine outfits across six requests, compared with 18 for quick and 14 for no memory: stronger personalization came with fewer alternatives. No model fallback was observed in 18 attempts, and hybrid document retrieval succeeded in 6/6 full runs. These denominators are too small to establish a production failure rate.

Evidence and interpretation

Astra rated the full retrieved evidence bundle 4.83/5 on usefulness across 12 ratings. Conditions without documents were assigned 1 by the rubric, so that contrast is not an independent retrieval-quality test. The judge received all retrieved sources; the generator’s bounded prompt used fewer passages. Source IDs are retained in the raw results so that distinction remains auditable.

For the first presentation brief, the judge identified a no-memory explanation that mentioned boots while its selected plan contained sneakers. Owned-ID validation prevents invented wardrobe records but does not guarantee grounded prose. The full condition respected the fictional red-color exclusion and preferred navy/sand combination; this can partly reflect deterministic filtering rather than better model reasoning.

Potential value, not demonstrated social impact

The findings support further testing of a more relevant owned-wardrobe assistant. They do not measure time saved, purchases avoided, waste reduction, human acceptance or correction burden. Astra is a same-provider-family, text-only evaluator with shared biases, not a fashion expert or a real participant. A small, feature-aligned synthetic study can overstate usefulness.

What do documents and memory actually add?

FOLLOW-UP / CONTROLLED ABLATION AND REPEAT LATENCY

Six new fictional briefs use the same 11 pieces and neutral explicit preferences. Each brief has a target top, bottom and shoes fixed before inference. Four conditions independently enable documents and feedback: neither, documents only, feedback only, and both. Feedback consists of three helpful votes for that trio. Condition order rotates; the both condition repeats immediately. These are 30 real OpenAI generation attempts, not simulated outputs.

Target hit@3 means at least one of the first up to three returned looks contains all three target IDs; extra layers are allowed. Failed or empty attempts are misses. This is a diagnostic preference-recall proxy, not human acceptance or fashion quality. Documents share six event-specific notes, three distractors and the standard guides; only the actual bounded prompt passages count as exposed evidence.

Initial attempts	Neither	Docs only	Feedback	Both
Target hit@3	1/6 (16.7%)	3/6 (50.0%)	3/6 (50.0%)	5/6 (83.3%)
Target in candidate plans	3/6	3/6	5/6	5/6
Median end-to-end*	3.43 s	3.52 s	3.88 s	3.14 s
Model fallback	0/6	0/6	0/6	0/6

Where the measured change occurs

Adding documents without feedback improved target matching on 2/6 briefs, worsened 0/6 and tied 4/6. With feedback already enabled, documents improved 2/6, worsened 0/6 and tied 4/6. This separates the two mechanisms within these authored cases; six cases and stochastic generation do not establish a population effect.

Failure: the design-critique target never entered the 12 candidate plans in any arm, despite all its pieces being retrieved and its note reaching the prompt. The model could not select that combination. Feedback expanded target-plan coverage from 3/6 to 5/6 overall, but did not solve this case. Candidate selection is therefore a higher priority than simply increasing model size.

The target note reached the actual generator prompt in 12/12 initial document-enabled calls. Lexical retrieval ranked the target first in 6/6 queries; hybrid ranked it first in 12/12 initial queries. Event names occur in both notes and briefs, so this diagnostic cannot establish a semantic-search advantage. All 54 returned outfits were checked; 54 passed ownership/completeness, with 0/30 model fallbacks and 0/30 wardrobe-retrieval fallbacks.

Cache benefit versus user-visible latency

Same six briefs	Initial / empty cache	Immediate repeat
Median retrieval	509 ms	231 ms
Median end-to-end*	3.14 s	2.60 s
Wardrobe vector cache hits	0/72	72/72
Target hit@3	5/6 (83.3%)	5/6 (83.3%)

Repeats were faster in 5/6 pairs; the median paired repeat-minus-initial difference was -0.51 s. *Timings cover local orchestration plus live provider calls, with prebuilt document indexes. They exclude Render idle startup, camera upload and human decision time. Exact prompts and attempts are in ablation.json.

A measurable alpha. A focused next iteration.

IMPLEMENTATION AND RELEASE STATUS

The app now includes a paired study workspace with blinded set labels, explicit consent, per-look ratings, evidence ratings and anonymized exports. Failed, cancelled and unscored attempts remain visible. Development rehearsals are excluded from human results. Photo review records active visible-tab review time and edited fields; it does not infer total task time or label correctness from an edit alone.

Hosted access uses private participant codes, signed HttpOnly/Secure cookies and server-derived UUID namespaces. Knowledge and photo-job ownership is checked on the server. Cross-account document and job tests use the real HTTP routes and SQLite store. The final regression suite passed 169/169 tests; the production frontend build passed. Browser inspection confirmed the study workspace renders without console warnings or errors.

Deployment: The full app is live on Render Free over HTTPS. All nine hosted smoke checks passed, including private sign-in, cross-account isolation, semantic document import, a real photo job and OpenAI RAG styling. The single hosted photo job took 5.52 s after enqueue; one styling request took 11.08 s and returned two valid owned outfits. These are deployment checks, not population latency estimates.

https://closet-quest-junhao-s7.onrender.com

The free Render service is an alpha host, not durable cloud storage [4]. It sleeps after inactivity and loses server-local SQLite records and uploaded job files on restart. Browser closets persist on that device; imported knowledge must be backed up separately or re-imported. OpenAI usage is billed separately. Shared-device confidentiality, account recovery, cross-device wardrobe sync and durable storage remain unfinished.

Priority	Observed reason	Next validation gate
1. Candidate coverage	The critique target was absent in all four arms; documents could not recover it.	Preserve preference-relevant combinations before model selection; rerun on new briefs.
2. Recognition and prose	One targeted photo still fails; valid IDs can coexist with incorrect advice.	Geometry recovery; fresh phone-photo labels; check garment mentions against selected records.
3. Real usefulness	Authored target recall and synthetic ratings do not establish user value.	Counterbalanced same-person briefs; top-three acceptance, evidence usefulness, decision time and correction burden.
4. Durable deployment	Free host storage is ephemeral; camera hardware unverified.	Persistent account-scoped database and recovery; physical-phone camera and hosted idle-start measurements.

Sources and reproducibility

[1] Fashionpedia dataset and annotation format. [2] Fashionpedia license: annotation mapping is adapted from CC BY 4.0 data; image licenses are recorded individually. [3] OpenAI model documentation, checked October 19, 2026. [4] Render free-service limitations.

Supporting results: output/sprint7 contains the fixed dataset manifest, per-image JSON/CSV, fictional briefs and judge reasons, aggregate metrics, targeted recheck, test log and the controlled ablation with exact prompts and cache counters. Reproduction scripts are under scripts/. Dataset photos and credentials are excluded from the repository.